Distillation vs RL

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In one line: RL is learning to cook by experimenting and tasting. Distillation is copying a great chef's recipe notebook.

How they differ

	RL (GRPO)	Distillation (SFT on traces)
Learns from	Its own attempts + a score	A teacher's finished answers
Needs	Problems with checkable answers	A stronger teacher, or a dataset of its traces
Signal	One number per attempt	A target for every token
Can beat the teacher?	Yes, in principle	Rarely
Fails when	The model never gets an answer right (all advantages are 0)	It copies the style (layout, length) without the skill

Words to know

- **Soft labels:** the teacher's full probabilities ("42: 0.82, 41: 0.11 ..."), not just the top answer (Hinton, Vinyals & Dean, 2015).
- **Temperature:** above 1 flattens soft labels so near-misses show; below 1 sharpens them.
- **Sequence-level distillation:** fine-tune on the teacher's written traces (what "distilled from R1" usually means). No teacher at runtime.

What the papers found (as reported)

DeepSeek-R1 (DeepSeek-AI, 2025): about 800k samples curated with R1 → SFT on Qwen (1.5B, 7B, 14B, 32B) and Llama (8B, 70B).

Qwen-32B base	AIME 2024	MATH-500	GPQA Diamond	LiveCodeBench
RL directly (R1-Zero-Qwen-32B)	47.0%	91.6%	55.0%	40.2%
Distilled from R1	72.6%	94.3%	62.1%	57.2%

Even R1-Distill-Qwen-1.5B reported 28.9% on AIME 2024 and 83.9% on MATH-500.

s1 (Muennighoff et al., 2025): 1,000 curated examples (s1K) on Qwen2.5-32B-Instruct; "budget forcing" (append "Wait" to think longer). **LIMO (Ye et al., 2025):** 817 curated examples in v1 (57.1% AIME, 94.8% MATH); a later version reports 63.3% AIME24, 95.6% MATH500. **Caution:** both used 32B models and heavily filtered data.

Which one, when?

Situation	Choose
Small model + good teacher traces exist	**Distill**
No teacher, answers easy to check, model sometimes right	**RL**
Model scores 0%	**Distill or make the task easier, then RL**
Need to beat the best existing model	**RL**
Distilled model is accurate but too long	**Distill, then RL with a length reward**

Default recipe for small models: distill first (learn the habit) → GRPO with a checkable reward (sharpen it) → watch length.

Lab settings (TRL 1.16.0, free T4)

`SFTTrainer` + `LoraConfig(r=16, target_modules="all-linear")` · `completion_only_loss=True` · `fp16=True, bf16=False` · `max_length=1024` · data: `camel-ai/gsm8k_distilled` (MIT) · test: GSM8K test split, greedy decoding.